

Problem 1: DNN and Wine Classification (80 points):

To reduce the model size, I combined knowledge distillation with INT 8 quantization. INT 8 quantization allowed for the model to be represented with 8-bit integers instead of float values. This reduced the storage size to about 3.68KB, which was the smallest model. I thought that this may result in decreased accuracy however, the model achieved about a 0.9630 test accuracy, with high precision, recall, and F-1 Score. The confusion matrix also showed very little misclassification.

Problem 2: Exploring Edge Impulse (20 points):

1. [Dis] (5 pts) What types of “Learning Blocks” are available under the free-tier Edge Impulse account (e.g., supervised learning—classification and regression, unsupervised learning—clustering, self- supervised learning, etc.)?

Edge Impulse includes several learning blocks in its free version, covering supervised, unsupervised, and transfer learning. For supervised tasks, Classification works well for building neural networks which can be applied to recognize keywords like "on," "off," or "yes" from a microphone. There is also Regression for predicting continuous numbers, which can be applied to estimate temperature from sensor data. Object Detection (using FOMO or MobileNetV2 SSD) handles localization and identification, useful to identify objects in pictures. Lastly, Transfer learning blocks let you fine tune pretrained models.

For unsupervised learning, the Anomaly Detection (GMM) block builds a Gaussian Mixture Model of "normal" behavior to flag anything unusual, like a motor vibrating incorrectly. The K-means version of Anomaly Detection clusters sensor data to find outliers, while Visual Anomaly Detection (FOMO-AD) brings this to images. This can be used to identify surface defects on parts coming off an assembly line.

For each type, provide at least one example model that can be implemented using the free tier.

2. [Dis] (5 pts) List all the types of layers that can be used under the “Classifier” section in the free-tier Edge Impulse. Briefly explain the purpose of each layer identified.

Layer	Purpose
Dense	Used to process feature data.
1D Convolution / pooling	Learns features that take spatial information into account along a single

	dimension.
2D Convolution/ pooling	Learn features that take spatial information into account along two dimensions.
Reshape	Transforms 1D data into multidimensional data.
Flatten	Transforms multidimensional data into 1D.
Dropout	Reduces model overfitting.

3. [Dis] (5 pts) What types of model compression options are available under the “Deployment” section in the free-tier Edge Impulse? List all available methods.

Under the “Deployment” section I only see quantization and different compiler options for model compression options.

Within the "Deployment" section, there are both quantization and compiler methods. For quantization methods we have INT8 Quantization and Float 32 quantization. INT8 Quantization reduces weights and activations down from 32-bit floats to 8-bit integers. That’s about a fourfold drop in size and a big speed boost for microcontrollers. With Float32 (Unoptimized), there’s a full 32-bit precision with no compression. However, this works well as a baseline for accuracy checks.

For compilers, there’s the EON Compiler which takes your model and transforms it into C++ code customized for your microcontroller. There’s also an EON Compiler (RAM Optimized) which recalculates intermediate activations instead of storing them, which reduces RAM usage.

4. [Dis] (5 pts) What input modalities are supported for synthetic data generation in the free-tier version of Edge Impulse? For each modality, list the synthetic data generation methods/algorithms provided. Briefly explain why synthetic data is important in machine learning applications

The free tier supports synthetic data generation for the following input modalities:

Images:

- Black Forest Labs Flux-Pro Image Generator
- DALL-E 3 Image Generators

Audio:

- Whisper Synthetic Voice Generator
- NVIDIA Omniverse

Synthetic Data is important because in the real world, obtaining real-world data sets that are properly labeled can be slow, costly, and time-consuming. Synthetic data streamlines this process because it generates as many labeled samples as needed. Additionally, it doesn't carry any privacy risks because there aren't real people involved.